

Analytical Problem Solving Activity

classmate

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1) Gradient Magnitude and Direction

$$G_x = 24 \quad ; \quad G_y = 18$$

$$\text{Gradient Magnitude} = \sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$$

$$\begin{aligned} \text{Gradient direction} = \theta &= \tan^{-1}\left(\frac{G_y}{G_x}\right) \\ &= \tan^{-1}\left(\frac{18}{24}\right) = \tan^{-1}(0.75) \\ &= 36.87^\circ \end{aligned}$$

2) Non-Maximum Suppression

Given data :-

Pixel	P1	P2	P3	P4	P5
Magnitude	25	50	90	60	30

→ P1(25) compare with P2(50)

$25 < 50 \rightarrow$ suppressed (0)

→ P2(50) compare with P3(90) and P1(25)

$50 < 90 \rightarrow$ suppressed (0)

→ P3(90) compare with P4(60) and P2(50)

$90 > 50$ and $90 > 60 \rightarrow$ keep

→ P4(60) compare with P3(90) and P5(30)

$60 < 90 \rightarrow$ suppressed (0)

→ P5(30) compare with P4(60)

$30 < 60 \rightarrow$ suppressed (0)

Final output :-

Pixel	P1	P2	P3	P4	P5
output	0	0	90	0	0

After NMS :- [0, 0, 90, 0, 0]

③ Double Thresholding :-

given

High Thresholding (T_h) = 100
low Threshold (T_l) = 50

pixel	P1	P2	P3	P4	P5	P6
magnitude	135	90	45	120	70	20

Rules for classification :-
 $C \geq 100$ - strong edge
 $50 \leq C < 100$ - weak
 $C < 50$ - Non-edge

$P1 = 135 \rightarrow \geq 100 \rightarrow$ strong edge

$P2 = 90 \rightarrow$ between 50 and 100 $\rightarrow$ weak edge

$P3 = 45 \rightarrow < 50 \rightarrow$ Non-edge

$P4 = 120 \rightarrow \geq 100 \rightarrow$ strong edge

$P5 = 70 \rightarrow$ between 50 and 100 $\rightarrow$ weak edge

$P6 = 20 \rightarrow < 50 \rightarrow$ Non-edge

Final Answer:-

Pixel	P1	P2	P3	P4	P5	P6
classification	strong	weak	Non edge	strong	weak	Non edge

④ given

Pixel	P1	P2	P3	P4	P5	P6	P7
magnitude	130	95	40	115	65	25	105

High Threshold = 100

low Threshold = 50

1. strong edges.

Magnitude ≥ 100

$P1 = 130$

$P4 = 115$

$P7 = 105$

Strong edges = $P1, P4, P7$

2. Weak Edges

$50 \leq \text{Magnitude} < 100$

$P2 = 95$

$P5 = 65$

Weak edges = $P2, P5$

3. Non edges:-

Magnitude < 50

$P3 = 40$

$P6 = 25$

Non edges = $P3, P6$

4. Final edge Pixels after Hysteresis

$P2 \rightarrow P1 = \text{retained}$

$P5$ not connected to strong edge - Removed

strong edges $P1, P4, P7 \rightarrow \text{retained}$

Final edge Pixels = $P1, P2, P4, P7$

House circle Transform

1. Given

Edge points $P_1 = (5, 4)$, $P_2 = (7, 6)$
Candidate centres $C_1 = (2, 4)$, $C_2 = (4, 3)$

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

1. Centre $C_1(2, 4)$

For $P_1 = (5, 4)$

$$r = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3$$

For $P_2 = (7, 6)$

$$r = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{29} = 5.39$$

$$C_1 = r = 3, 5.39$$

$\therefore$ No matches

2. Centre $C_2(4, 3)$

For $P_1(5, 4)$

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2} = 1.4$$

For $P_2(7, 6)$

$$r = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$$

$$C_2 = r = 1.4, 4.24$$

$\therefore$ No matches

2) Given

edge point $P = (8, 6)$

candidate centre $C = (4, 3)$

$$\begin{aligned} r &= \sqrt{(x-h)^2 + (y-k)^2} \\ &= \sqrt{(8-4)^2 + (6-3)^2} \\ &= \sqrt{4^2 + 3^2} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

3) given

Centre $C = (5, 4)$

Edge point $P = (8, 8)$

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8 - 5)^2 + (8 - 4)^2}$$

$$= \sqrt{3^2 + 4^2} = \sqrt{25}$$

$$= 5$$